INTERNAL HOM OUT OF A DERIVED COLIMIT OF PERFECT OBJECTS IS THE DERIVED LIMIT OF THE TENSOR PRODUCTS WITH THE DUALS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the rewriting of the tensor products with the duals as internal homs, the two displayed triangles, the reduction to the product identity, and the formal verification from the internal hom equation.

Editorial note. The label lemma-colim-and-lim-of-duals names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement retains the author's slogan environment (a comment environment in the chapter preamble), delivered verbatim; the three cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. Because the counted statement carries the author slogan environment, the preamble excerpt includes the shared Stacks Project definitions of the comment environments (verbatim package plus reference, slogan and history), taken verbatim from the chapter preamble excerpt of the same project snapshot.

Allowed standard prerequisites (editorial).

- The derived dual of a perfect complex and its functorial properties. Same chapter, the lemma describing the dual of a perfect object and the identification M tensor^L K^vee = RSheafHom(K, M). Supplies the rewriting of R lim E tensor^L K_n^vee as R lim RSheafHom(K_n, E).
- The defining triangles of derived limits and colimits. Derived Categories, the definition of the derived colimit, giving the triangles R lim RSheafHom(K_n, E) -> prod -> prod and bigoplus K_n -> bigoplus K_n -> K. Supplies the two triangles used to reduce to a single formal identity.
- The internal hom equation and commutation of the tensor product with direct sums. Same chapter, the equation describing the internal hom complex and the commutation of the derived tensor product with direct sums. Supplies the formal identification RSheafHom(bigoplus K_n, E) = prod RSheafHom(K_n, E).

Source reference. Original definition 090Z: sources/stacks-chow/tags.txt, source lines 11427–11427. Prerequisite pointer invoked at proof line(s) 13423; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:sites-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original equation 08DI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 8844–8844. Prerequisite pointer invoked at proof line(s) 13444; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 07VI: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 13048–13048. Prerequisite pointer invoked at proof line(s) 13432; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $(K_n)_{n\in\mathbf{N}}$ be a system of perfect objects of $D(\mathcal{O})$. Let $K = hocolim K_n$ be the derived colimit (Derived Categories, Definition 090Z). Then for any object E of $D(\mathcal{O})$ we have*

$$R\mathcal{H}om(K, E) = R\lim E \otimes^{\mathbf{L}}_{\mathcal{O}} K_n^\vee$$

where $(K_n^\vee)$ is the inverse system of dual perfect complexes.

Proof. By Lemma 07VI we have $R\lim E \otimes^{\mathbf{L}}_{\mathcal{O}} K_n^\vee = R\lim R\mathcal{H}om(K_n, E)$ which fits into the distinguished triangle

$$R\lim R\mathcal{H}om(K_n, E) \to \prod R\mathcal{H}om(K_n, E) \to \prod R\mathcal{H}om(K_n, E)$$

Because K similarly fits into the distinguished triangle $\bigoplus K_n \to \bigoplus K_n \to K$ it suffices to show that $\prod R\mathcal{H}om(K_n, E) = R\mathcal{H}om(\bigoplus K_n, E)$. This is a formal consequence of (08DI) and the fact that derived tensor product commutes with direct sums. □